

Maham Kamran

Computational Fluid Dynamics Research Engineer — MSc Mechanical Engineering

Portfolio | [LinkedIn](#) | +1 226 338 6621 | mkamran2@lakeheadu.ca | Thunder Bay, Canada

EXPERIENCE

R&D Research Associate: Computational Fluid Dynamics Engineer

June 2025 – Present

Nature-Inspired Engineering Research Lab (NIERL)

Thunder Bay, Canada

- Designing and developing a bio-inspired fin propulsive mechanism for marine vessels, integrating mechanical design with fish-like kinematics.
- Performing high-fidelity CFD analyses to study hydrodynamic performance, optimizing fin geometry and motion for improved propulsion efficiency and maneuverability.

R&D Engineer: Graduate Research Assistant [Publication: [Link](#), Preprint: [Link](#)]

September 2023 – May 2025

Lakehead University (NIERL-Research Lab)

Thunder Bay, Canada

- Conducted a comprehensive 3D investigation of integrated vortex–odor dynamics for solitary undulatory swimmers.
- Performed quantitative analysis of normalized odor effectiveness to assess overall odor detection performance.
- Executed CFD simulations of odor–vortex dynamics using an immersed-boundary solver, focusing on waveform patterns, Strouhal and Reynolds numbers, and diffusion-convection interplay for bio-inspired AUV design.

Research Fellowship | MITACS Internship

January 2025 – March 2025

CWRU: Flow Simulation & Flow Physics Lab

Cleveland, USA

- Conducting comprehensive research on Fluid–Structure–Chemical Interactions (FSCI) in 3D undulatory swimmers, with emphasis on schooling behavior in anguilliform and carangiform species.
- Leveraging advanced CFD simulations to investigate coupled vortex–odor dynamics in fish wakes for improved underwater sensing and bio-inspired AUV design.

Graduate Asset Integrity Engineer

August 2022 – July 2023

ABYSS Solutions Inc

Islamabad, Pakistan

- Inspected offshore oil and gas platforms for Occidental-Anadarko and British Petroleum using P&ID drawings and Navisworks GAD models.
- Supervised development and deployment of machine learning workflows for contextualizing platform layouts and detecting corrosion on pressure equipment for Abyss Fabric products, leveraging dense cloud tagging for enhanced analysis.
- Performed discrepancy analysis and redlining of P&IDs to improve offshore asset integrity, reduce operational risk, and support safer workplaces.
- Prepared corrosion assessment and condition monitoring reports in accordance with API standards, including API 580 (Risk-Based Inspection), API 570 (Piping Inspector), and API 571 (Corrosion and Materials).

Analytical Aero-thermal Engineer

September 2021 – June 2022

Shock and Stars Pvt Ltd

Islamabad, Pakistan

- Confidential project (NDA). Conducted comprehensive research on aerothermal behavior and cooling strategies for electronic enclosures, ensuring compliance with MIL-STD-5400 requirements for line-replaceable units (LRUs) across varying altitudes.
- Introduced and evaluated multiple passive and active heat-transfer methods within the aerothermal design framework.
- Developed an analytical aerothermal model and implemented solution workflows.
- Validated analytical and experimental findings through detailed Computational Fluid Dynamics (CFD) simulations.

Production Engineer Intern

August 2021 – September 2021

Oil and Gas Development Company Limited (OGDCL)

Islamabad, Pakistan

- Monitored performance for directional drilling operations in onshore oil and gas production facilities.
- Analyzed various units, including Amine, Dehydration, LPG, and Condensate Extraction Units, to assess operational efficiency.
- Identified and evaluated different systems in operations, including Ignition, Fuel, Lubrication, and Cooling systems.

EDUCATION

Lakehead University | Masters of Science (MSc), Mechanical Engineering (**CGPA: A+**) Thunder Bay, Canada
Thesis: Computational Investigations of Integrated Vortex-Odor Dynamics [Video](#), [Publication](#) Sept. 2023 – May 2025
Courses: Nonlinear Vibrations and Chaos | Turbulence Modeling | Computational Mechanics | Turbomachinery
Advisor(s): [Dr. Saif Khalid](#)

Air University | Bachelor of Science (BSc), Mechanical Engineering (**CGPA: A+**) Islamabad, Pakistan
Thesis: Aerothermal Design Analysis of Dorsal Mounted Avionics LRU in Unconditioned Bay Sept. 2018 – Aug. 2022

SKILLS

Languages: C | C++ | Python | Object-Oriented Programming | SSCNC | MATLAB (Simulink)

Software: ANSYS | SolidWorks | AutoCAD-Autodesk | OpenFOAM

Inspection Standards: Asset Integrity | API 570 (Dead legs) | API 571 (Corrosion and Materials) | API 510 (Pressure Vessel)

ACHIEVEMENTS

- Awarded Graduate Student (Best) President's Award at Lakehead University, Canada.
- Awarded Oral Presentation Award at the Science and Environmental Studies Biotechnology and Allied Sciences Symposium (SESBASS)
- Awarded a 100% Undergraduate Higher Education Commission (HEC) Scholarship at Air University, Pakistan.
- Completed an Industrial project sponsored by Air Headquarters, Pakistan.

RESEARCH PUBLICATIONS

(J2) **M. Kamran**, A. Fardi, C. Li, and M. S. U. Khalid, "On the role of morphology, kinematics, and spatial arrangements of biological swimmers for dynamics of odors in the wake," 2025. [\[Preprint\]](#)

(J1) **M. Kamran**, A. Fardi, C. Li, and M. S. U. Khalid, "How does vortex dynamics help undulating bodies spread odor?," *Physics of Fluids*, 2024. [\[DOI\]](#)

PEER-REVIEWED CONFERENCE PAPER AND PRESENTATION

(C3) Z. Maleksabit, **M. Kamran**, A. Tarokh, and M. S. U. Khalid, "Examining the vortex dynamics from burst-and-coast motion of an anguilliform swimmer," abstract submitted to the *14th International Symposium on Turbulence and Shear Flow Phenomena (TSFP13)*, Heidelberg, Germany, 2026.

(C2) **M. Kamran**, C. Li, and M. S. U. Khalid, "On the role of physiology and kinematics of biological swimmers to spread and suppress their odors in the wake," *CSME-CFDSC-CSR 2025 International Congress*, Montreal, QC, Canada, 2025.

(C1) **M. Kamran** and M. S. U. Khalid, "Coupled vortex-odor interactions in fish swimming for underwater sensing," *Science and Environmental Studies Biotechnology & Allied Sciences Symposium*, Thunder Bay, ON, Canada, 2025.

POSTER PRESENTATION

(P1) **M. Kamran**, C. Li, and M. S. U. Khalid, "How do fish spread odor in their wake? A perspective from Computational Fluid Dynamics," *Professional Engineers Ontario (PEO) Lakehead Chapter Technical Conference*, Thunder Bay, ON, Canada, 2024.

ACADEMIC EXPERIENCE

Peer-reviewed research publication for American Institute of Physics (AIP)

Journal: Physics of Fluids, ISSN: 1089-7666

Canada
October, 2024

Teaching Assistant, Lakehead University

EMEC-0538 – Feedback Control Systems

Canada

Fall 2024

EMEC-3454 – Applied Heat Transfer

Winter 2024

EMEC-3436 – Engineering Thermodynamics

Fall 2023